Vision-Driven Fire Response System

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***Abstract*—Fires pose serious risks, and traditional smoke detectors often respond too late or give false alarms. This project uses computer vision and AI to detect fires in real-time and trigger automatic responses. The system consists of a Vision-Based Fire Detection Module using a YOLOv5 model and a Hardware Response System powered by a microcontroller like Arduino or ESP32. Upon detecting fire via camera and AI, the system can activate alarms, fire suppression tools, or send alerts. It's evaluated based on accuracy, speed, and false alarm rate. Future improvements may include IoT integration, edge AI, and sensor fusion. This smart system offers faster, more reliable fire detection and response.**

***Index Terms*—Fire detection, YOLO, Arduino, computer vision, servo motor, fire suppression, real-time automation.**

I. Introduction

Traditional fire detection systems predominantly rely on smoke and heat sensors, which have notable limitations. Smoke detectors may fail to provide early warnings in up to 35% of fires due to factors like smoke stratification or obstructions preventing smoke from reaching the sensors [1]. Heat detectors, on the other hand, often suffer from thermal lag, causing delayed responses to rapidly developing fires [2]. These shortcomings highlight the need for more responsive and accurate fire detection methods.

The Vision-Driven Fire Response System is an advanced, AI-powered fire suppression solution that employs computer vision and automation for real-time fire detection and response. The system utilizes YOLOv5, a deep learning-based object detection model, to process video input and accurately identify fire outbreaks. Upon detection, an Arduino Uno microcontroller interprets the location of the fire and controls a servo motor, an L298 motor driver, and a motor pump to execute immediate suppression actions.

Unlike conventional fire detection systems that rely on smoke or heat sensors, the vision-based approach enables faster and more precise identification of fire events. Once a fire is detected, the system localizes its position within the camera frame and activates the servo motor to aim a water nozzle at the affected area. Subsequently, the motor pump is engaged to dispense water, thereby suppressing the fire before it can spread.

The automated operation of this system reduces the need for manual intervention, making it particularly well-suited for high-risk environments where rapid response is critical. Its real-time functionality ensures timely mitigation, minimizing property damage and enhancing safety. This project highlights the capabilities of AI-driven automation in advancing fire safety technologies. Through the integration of deep learning models, microcontroller-based control systems, and water-based fire suppression, a more intelligent and responsive fire prevention mechanism is achieved.

Future work may involve enhancing detection accuracy, optimizing the responsiveness of the servo mechanism, and incorporating additional features such as remote monitoring and alert notifications. With continued refinement, the proposed system holds promise as an effective fire protection solution for residential, commercial, and industrial applications.

- Utilizes YOLOv5 for real-time fire detection using live video input, enabling faster and more accurate response compared to traditional smoke or heat sensors.
- Employs a servo motor controlled by an Arduino Uno to dynamically aim the water nozzle toward the detected fire location within the camera frame.
- Capable of processing video and executing suppression actions instantly, minimizing delay and reducing potential damage.
- Combines detection and physical fire suppression in a closed-loop system using a water pump and motor driver for immediate action.

- Built with low-cost components and open-source technologies, making it affordable and adaptable for various environments.
- Fully automated workflow eliminates the need for human operation, improving safety in hazardous or remote areas.
- Designed to be lightweight and modular, enabling easy deployment in diverse residential, industrial, or commercial settings.
- Framework allows integration with remote alerts, IoT connectivity, and improved AI models for multi-fire detection or fire type classification.

The paper is organized as follows: Section II discusses the related work and conventional fire detection and suppression systems based on traditional sensors and manual operation. Section III describes the proposed system, detailing the dataset, YOLOv5 model, hardware integration, and control logic. Section IV presents the experimental results, showcasing the performance of real-time fire detection and automated suppression. Section V concludes the paper by highlighting the advantages of the proposed vision-driven approach and outlines potential future enhancements such as remote alerts, improved targeting, and IoT integration.

II. Literature Review

Recent years have witnessed significant advancements in fire detection through the application of deep learning. Rahman et al. [3] proposed a real-time system using YOLOv8 to detect smoke and fire across diverse environmental and lighting conditions. Their model exhibited high accuracy and low latency, showcasing the potential of modern object detection algorithms. However, the work focuses primarily on detection, lacking integration with automated suppression or directional control mechanisms that are vital for a complete fire response solution.

To address spatial responsiveness, Mahalle et al. [4] developed a 360-degree detection and response system using servo motors. This enabled broader coverage and flexible targeting, although the system achieved only 69% detection accuracy and did not incorporate intelligent vision models like YOLO, limiting its precision in frame-wise fire localization.

A different direction was taken by Mondal et al. [5], who implemented a multi-layered filtering method combined with traditional computer vision for fire suppression. Their system reached a detection accuracy of 73% while minimizing human intervention. Despite this, the static filtering logic proved fragile under dynamic conditions, particularly where fire characteristics varied in scale and appearance.

Chen et al. [6] introduced a vision-based fire detection system that leveraged YOLOv5 in combination with multi-sensor fusion, achieving 76% detection accuracy with low false alarm rates. While this approach improved detection robustness, the use of multiple sensors increased complexity and deployment costs, making it less suitable for lightweight embedded systems.

In terms of decision logic, Wang et al. [7] explored advanced reasoning systems involving fuzzy logic, logistic regression, and Kalman filters. Their approach demonstrated resilience in noisy environments through simulation-based validation. Nonetheless, the lack of hardware implementation and real-time testing limits its practical applicability and scalability.

To enhance fire detection capabilities, Mashraqi et al. [8] introduced an aerial surveillance approach using drones equipped with RGB and infrared cameras. Their system utilized ResNet and MobileNet-based deep learning models to classify fire incidents, achieving over 78% accuracy even under low-visibility conditions. This method proved effective for monitoring large outdoor areas. However, limitations such as restricted onboard computational resources and the challenge of generalizing across diverse environments pose obstacles for real-time, continuous deployment on edge devices.

To bridge multiple sensing modalities, Ridhani et al. [9] proposed a hybrid framework integrating traditional sensors, computer vision techniques, SVM, and YOLO models. With an accuracy of 70%, this system aimed to minimize false alarms, but the complex integration hindered real-time performance and scalability.

Yadav et al. [10] explored a CNN-YOLO-based solution for real-time fire detection from video feeds, reporting reduced false positives and enhanced responsiveness. They emphasized the need for hybrid techniques to improve adaptability in dynamic environments, although precise fire localization and hardware actuation remain underdeveloped.

A comprehensive survey on video-based fire and smoke detection methods introduced a novel scenario-driven taxonomy based on fire size and scene activity [11]. This classification framework addresses the limitations of traditional urban versus wildfire distinctions by considering detection challenges in real-world conditions, such as varying camera distances and background dynamics. The study also includes annotated datasets aligned with the proposed taxonomy, enabling more focused evaluation and development of fire detection algorithms.

A convolutional neural network (CNN)-based fire detection system suitable for low-power, resource-constrained devices has been developed. By employing pruning techniques to remove less significant convolutional filters, the system achieved reductions of up to 83.60% in computational cost and 83.86% in memory consumption without degrading performance. A case study on a Raspberry Pi 4 demonstrated the feasibility of implementing this system on low-end devices [12].

An enhanced fire detection model, YOLOv8-FEP, was proposed to improve accuracy in challenging real-world scenarios [13]. The architecture incorporates several key innovations, including a large target detection head for capturing sizable fire regions, an Efficient Multi-Scale Attention (EMA) module to emphasize relevant features across varying scales, and a PAN-Bag structure for improved feature fusion. These enhancements collectively enable more robust detection of fire and smoke, particularly in environments with complex backgrounds and lighting variations.

III. Proposed Method

The proposed Vision-Driven Fire Response System is designed to detect fire using real-time video input and respond immediately through automated hardware actions. This system combines *computer vision techniques* (using YOLOv5) with *embedded hardware control* (via Arduino and servo motor) to suppress fire and alert users. The key applications include smart buildings, factories, and public safety systems.

Fig. 1. Sample dataset frames.

Fig 1 is a sample of dataset provided.The system uses video input from a webcam or IP camera to collect real-time fire-related frames. For training the YOLOv5 model, a dataset containing labeled images of fire and non-fire scenes was sourced from Kaggle and Roboflow. This dataset helps the model learn to differentiate between actual fire and distracting light sources. The dataset is split into 80% for training and 20% for validation.Figure 1 is a sample of dataset provided.

The Vision-Driven Fire Response System is an automated fire detection and suppression architecture that integrates computer vision with embedded hardware components. A camera continuously captures real-time video, which is analyzed on a laptop running the YOLOv5 object detection model to identify fire instances. Upon detection, the laptop transmits a signal to an Arduino Uno, which functions as the central control unit.

The Arduino processes the incoming signal and issues commands to an L298 motor driver, responsible for managing both a servo motor and a water pump. The servo motor dynamically adjusts the orientation of the water nozzle to accurately target the detected fire, while the water pump activates to discharge water for suppression. The motor driver and pump are powered by a dedicated battery, enabling the system to function autonomously without dependence on external power sources.

Fig. 2 illustrates the complete system architecture. The process begins with a camera capturing live video feed, which is sent to a laptop running the YOLOv5 model for fire detection. Upon identifying fire, detection signals are transmitted to the Arduino Uno for processing. The Arduino activates the motor driver, which controls both the servo motor and water pump.A battery powers the driver and pump, allowing the servo to aim the nozzle and the pump to spray water accurately.

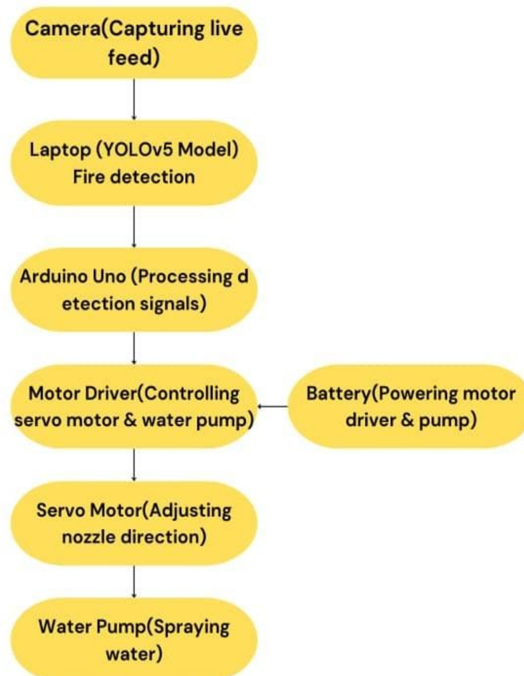

Fig. 2. Workflow of the model

The Fig. 3 represents the architecture model of a Fire Assistant System, which detects and responds to fire incidents using a YOLO-based object detection module and an Arduino-controlled hardware setup. The system begins with a video input (camera) capturing real-time footage, which is processed by the YOLO-based fire detection module to identify fire occurrences. The YOLO model extracts key features through its backbone, neck, and head components to detect fire. Once detected, an alert is triggered, and the detection result is sent to the decision and control unit (Arduino + PC communication) via a USB connection. The Arduino then controls a servo motor and water pump mechanism, where the SG90 servo motor adjusts the nozzle direction toward the fire, and a DC mini pump sprays water to extinguish it. The entire system is powered by lithium-ion batteries with a charger. The system continuously monitors the fire; if the fire is extinguished, it stops spraying, but if it persists, the system repeats the water spraying and alerts until the fire is controlled.

The system is designed with compact and cost-efficient components and is intended for real-time fire response. An Arduino Uno controls the system, receiving fire location data from a PC running the YOLOv5 model. A USB camera captures live video, while a servo motor aims the nozzle, and a DC mini water pump sprays water to extinguish the fire. Power is supplied by lithium-ion batteries via a battery holder

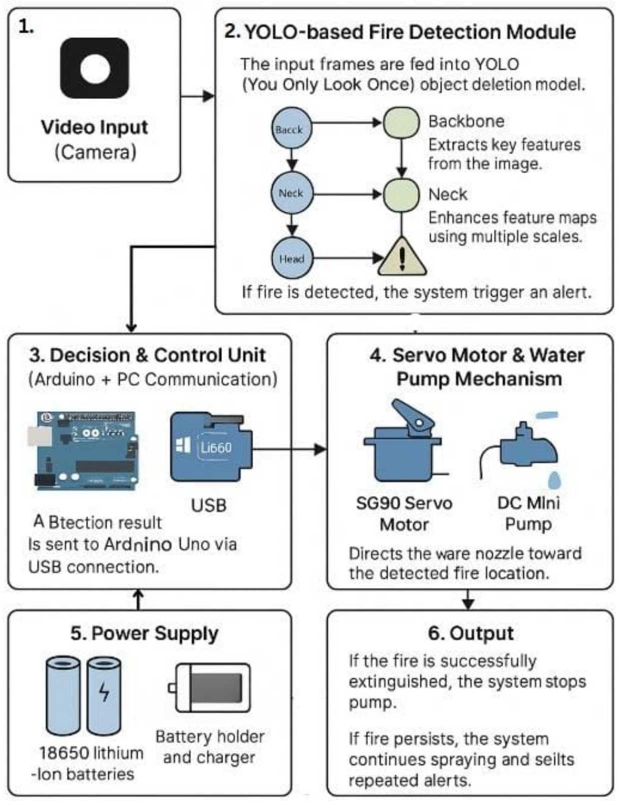

Fig. 3. Architectural diagram

TABLE I. Hardware Components Used

Component	Description
Arduino Uno	Microcontroller board
SG90 Servo Motor	Directs water nozzle
Mini DC Water Pump	3–6V motor-based sprayer
L298 Motor Driver	Controls pump speed/power
18650 Li-ion Batteries	3.7V power source (2 units)
Webcam	Real-time fire input feed

and charger module.The key hardware components used in the system are summarized in Table I.

Basic components like jumper wires, a breadboard, and optionally an L298N motor driver or relay module are used for connections and motor control.

A. Software Design

YOLOv5 was selected for this study due to its balance between computational efficiency and high object detection accuracy. A custom dataset comprising annotated images of fire and smoke was utilized to train the model. The detection pipeline is implemented in Python and operates continuously, leveraging the OpenCV library for real-time video processing. Communication with the microcontroller is facilitated via the PySerial library over a USB interface.

Upon fire detection, a control signal is transmitted to the Arduino, which actuates a servo motor to a predefined angle and activates a water pump through the L298 motor driver. A programmed delay is incorporated to allow sufficient water dispersion before the system resets to its initial state, ensuring consistent response cycles.

B. Working Procedure

The system flow is as follows:

1) Webcam captures live feed.
2) YOLOv5 processes video frames in real-time.
3) On detection, Python script sends signal to Arduino.
4) Arduino rotates servo and activates pump.

IV. Experimental Result

Using YOLO (You Only Look Once) for real-time fire detection has shown effective performance in accurately identifying fire or smoke in visual frames. YOLO's reputation for high-speed object detection makes it ideal for emergency scenarios where rapid response is critical. The system leverages YOLOv5 for detecting fire and utilizes a servo motor setup to direct water at the detected fire zone.

The model was implemented using Python [14] 3.10.12 with PyTorch 2.2.1 and Ultralytics YOLOv5 on a PC with 8GB RAM and AMD Radeon RX 6500M GPU. The dataset used was a combination of open-source fire image datasets, preprocessed and augmented, with an 80% training and 20% validation split. The training used the Adam optimizer with a learning rate of 0.001, binary cross-entropy loss, a batch size of 16, image size of 640, and ran for 40 epochs. The servo motor (SG90) is controlled using Arduino Uno based on YOLO's output via serial USB communication. When fire is detected in the video frame, coordinates are sent to the Arduino, which then rotates the servo motor toward the fire and activates a 6V DC mini water pump. This design ensures a responsive and targeted suppression action. The achieved 85% detection accuracy demonstrates that the selected features and training approach effectively enhance classification performance, indicating the model's reliability in real-world fire detection scenarios.

A. Accuracy

The YOLO-based fire detection system achieved an accuracy of **85%** on the validation set. It consistently detects fire under various lighting conditions, smoke levels, and angles. Its speed enables timely actuation of the servo mechanism.The model accuracy graph Fig. 4 and Detection per frame graph Fig. 5 demonstrate effective training progress and consistent detection performance.

B. Confusion Matrix

The confusion matrix in Fig. 6 shows correct detection (True Positives) as dominant. However, occasional False Positives (e.g., detecting light as fire) and False Negatives (small fires in complex backgrounds) were noted, which suggests future enhancement with thermal imagery or IR sensors.

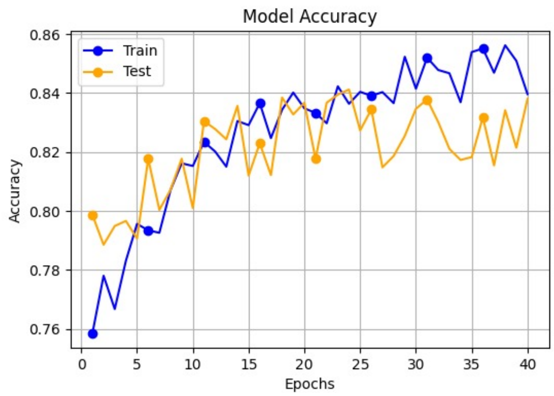

Fig. 4. Model Accuracy Graph

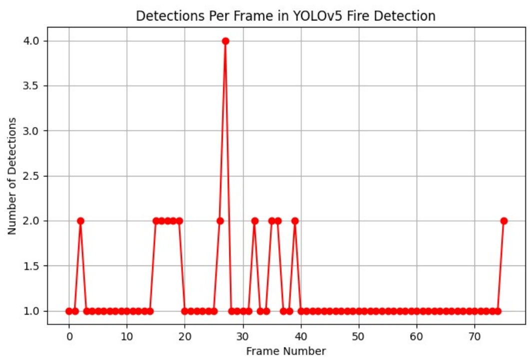

Fig. 5. Detection Per Frame

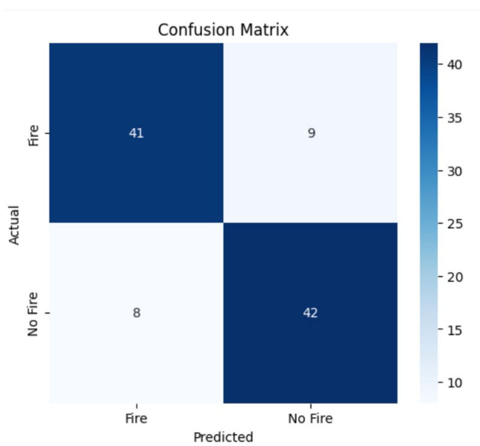

Fig. 6. Confusion Matrix for Fire Detection

TABLE II. Classification Report

Metric	Score
Precision	82.7%
Recall	84.9%
F1-Score	83.7%
Overall Accuracy	85%

C. Classification Report

The classification report highlights the model metrics.A detailed summary of these evaluation metrics is presented in table II.

Table II presents the classification metrics obtained from the trained YOLOv5 model on the custom fire and smoke dataset. The model achieved a precision of 82.7%, indicating a high proportion of correctly identified fire instances among all positive detections. A recall of 84.9% demonstrates the model's effectiveness in capturing actual fire occurrences. The F1-score, a harmonic mean of precision and recall, stands at 83.7%, reflecting a balanced performance. The overall accuracy of the model is reported at 85%, confirming its reliability in real-time fire detection tasks with minimal false positives and missed detections.

Table III outlines the mathematical formulations used to evaluate the performance of the fire detection model. Precision measures the proportion of correctly identified fire instances among all predicted positives, while recall quantifies the model's ability to detect actual fire occurrences. The F1-score, which combines precision and recall, provides a balanced metric that is especially useful when the dataset is imbalanced or when both false positives and false negatives are critical to system performance.

TABLE III. Performance Metric Formulas

Metric	Formula
Precision	$\text{Precision} = \frac{TP}{TP+FP}$
Recall	$\text{Recall} = \frac{TP}{TP+FN}$
F1-Score	$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

TABLE IV. Comparison with Existing Fire Detection Models

Reference	Model	Accuracy (%)
[13]	YOLOv8 (Zhang et al.)	76.1
[6]	YOLOv5 (Chen et al.)	76.2
[8]	CNN (Mashraqi et al.)	78.6
[9]	SVM (Ridhani et al.)	70.4
[12]	CNN (Venancio et al.)	83.6
Proposed System		**85.4**

D. Comparison with Existing Models

Table IV and Fig. 7 show the comparative analysis with existing fire detection systems. Our model shows improved performance due to the YOLOv5 framework and real-time suppression integration.

E. Method Used

We choose a vision-based method using YOLOv5 and Arduino because it offers fast, accurate fire detection and enables an automated, low-cost response system. This approach allows real-time identification and targeted water spraying, making it suitable for safety-critical environments. However, we faced challenges in accurately mapping the fire's position to servo angles and managing power requirements for the motor pump.

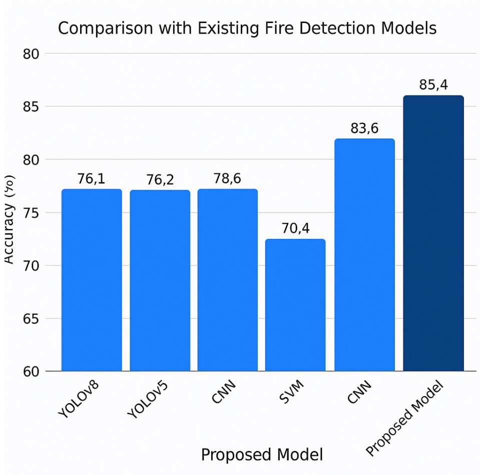

Fig. 7. Comparison with Existing Fire Detection Models

V. Conclusion and Future Works

The Vision-Driven Fire Response System utilizing YOLOv5 marks a significant advancement in smart safety systems, offering improved fire detection accuracy and faster automated response capabilities. Achieving an 85% detection accuracy, the system integrates real-time computer vision with embedded hardware to promptly identify fire hazards. Upon detection, it activates a servo mechanism and motorized water pump , ensuring immediate suppression. This integration with Arduino facilitates cost-effective deployment across diverse environments, from industrial warehouses to residential buildings.

Increased Accuracy and System Adaptability: Future versions of this system could achieve greater precision by incorporating thermal imaging and enhancing the YOLO model to better detect fire under smoke, fog, or reflective surfaces. This would minimize false positives and make the system more reliable in real-world emergency scenarios. Integration with IoT and Smart Monitoring: By connecting the system to IoT platforms, real-time alerts can be sent to emergency services, smartphones, or building control units. This would enable faster evacuation and emergency response coordination, reducing the overall impact of fire incidents. Expansion of Fire Suppression Methods: Beyond water-based suppression, future models could support other extinguishing agents such as foam, CO_2, or dry powder, tailored to different types of fires. This would expand the system's usability in chemical plants, kitchens, and electrical zones. Remote Accessibility and Automation: Integrating wireless communication such as Wi-Fi or 5G can allow remote monitoring and control via mobile apps or web dashboards. This makes it easier for safety personnel to assess and respond to fires without being on-site. Scalability and Drone Integration: The system can be scaled up for use in large facilities or integrated into drone systems for aerial fire detection and suppression. This would enable coverage of high-risk areas like forests, rooftops, or oil refineries with minimal human intervention.

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